

Make Your Own PBX with Asterisk

A Collage of Multiple Hardware Equipment



Asterisk is a software switching platform, capable of running on standardised PC hardware with accessories to connect to various telecom networks. In this second article in the series, the author describes various kinds of hardware and accessories that are needed to connect to different kinds of networks.

To understand and use the full power of Asterisk, we need to have a good understanding of the possible hardware accessories involved. The objective of this article is to introduce you to different kinds of hardware in the Asterisk world.

Nowadays, any standard desktop or server comes with IP connectivity. So, if we are connecting IP phones to each other through the PBX, we need not use any additional hardware. However, we also need to connect to landlines, mobile phones, etc. For this, there is hardware to connect to a PSTN network or a GSM network. Let us examine these devices, individually.

VoIP phones

VoIP phones are directly connected to the server over a LAN. The dominating protocol standard is SIP (Session Initiation Protocol). Some phones support IAX, a proprietary Asterisk

protocol, which uses less bandwidth and has better jitter buffer control to improve the audio quality. When starting, the phones boot themselves and register with the configured server and from then on, they can receive or make calls. The phones are also capable of registering with multiple servers, so that in redundancy scenarios, they can switch over to the redundancy server with ease.

Headsets turn PCs into VoIP phones

Free software like X-lite/Eyebeam or open source software like Blink can be downloaded and installed on a desktop. The GUI resembles a dial pad, and with the help of a headset with a microphone, the desktop can be turned into a phone. You may also use the microphone and speaker of the desktop or laptop. After starting the software, it registers itself with the Asterisk PBX and starts receiving as well as making calls.

Smartphones as VoIP phones

VoIP clients, available on smartphones, can connect to the PBX server over Wi-Fi and function like any other phone. One open source VoIP smartphone is the CSipClient, which can be downloaded freely from the Google Android Play Store. This client opens up innumerable application possibilities, as follows.

Phoneless desktop: Users need not clutter their desks with standard phones. Their mobile phones can serve as an